

MSCF Probability: Problem Session #1

You should complete as much as you can, but I believe that completing six questions is reasonable for the two hour period. You should not stop, however, if you have completed six exercises. You can choose to work on any exercises that you wish, but it's in your interest to make sure you cover a range of topics.

1. Show that $P(A | B) = 1 - P(A^c | B)$. You can assume that $P(B) > 0$.
2. A smoke detector system uses two devices, A and B . When smoke is present,
 - the probability that it is detected by device A is .85;
 - the probability that it is detected by device B is .90;
 - there is 11% chance that it is detected by *exactly one* of the devices.
 - (a) What is the probability that when smoke is present, it is detected by the system (i.e., by at least one of the two detectors).
 - (b) If smoke is *not* detected by device B , how likely is it to be detected by device A ?
 - (c) Are the two devices detecting smoke independently of each other?
3. In answering a question on multiple choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer, and assume that each question has m multiple-choice alternatives (only one of which is correct). Find an expression (involving p and m) for the conditional probability that a student actually *knew* the answer to a question, given that he or she answered it correctly.
4. Suppose that events A_1 and A_2 are independent conditional on event B . Show that $P(A_2|A_1 \cap B) = P(A_2|B)$. You can assume that $P(A_1 \cap B) > 0$.
5. Let $X \sim \text{Geometric}(p)$.
Prove that X has the *memoryless property*: For $n = 0, 1, \dots$ and $m = 0, 1, 2, \dots$, it holds that
$$P(X \geq n + m | X \geq n) = P(X \geq m)$$
6. A coin with $P(\text{Heads}) = p$ is tossed repeatedly. Find the probability that the first “Heads” occurs on an odd-numbered toss, and show that this probability is *always* more than $1/2$ (regardless of the value of p).
7. The number of bacteria in a 10 ml sample of swimming-pool water, X , has a Poisson distribution with mean $\lambda = 1.4$. Suppose that such a 10 ml sample of pool water is declared contaminated if at least two bacteria are found.

If 20 independent 10 ml samples of pool water are obtained, what is the probability that exactly half of them are contaminated?
8. Let the cumulative distribution function of a continuous random variable X be:

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ 1 - \sum_{k=0}^2 \frac{(2x)^k e^{-2x}}{k!} & x > 0 \end{cases}$$

Determine the pdf of X , $f_X(x)$.

9. Imagine a pair of independent Poisson processes. Events of “Type 1” arrive as a Poisson process with rate λ_1 . Events of “Type 2” arrive as a Poisson process with rate λ_2 . Now consider the “combined” process that results from ignoring the “type” of the events. The combined process is still a Poisson process. What is its rate? Explain your answer clearly.

10. The random variables X and Y have a joint density function given by

$$f_{XY}(x, y) = \begin{cases} 2e^{-2y}/y & 0 < y, 0 < x < y \\ 0 & \text{otherwise} \end{cases}$$

Compute $\text{Cov}(X, Y)$

Guidance: To find $E(X)$, note that finding $f_X(x)$ is not so “friendly”... Instead, consider $g(X, Y) = X$ and setup the $\int \int$ in a “convenient” way.

11. **(More challenging.)** Let $X \sim \text{Exp}(\lambda_1)$ and that $Y \sim \text{Exp}(\lambda_2)$.

(a) For $t > 0$, prove that $P(Y < tX) = \frac{t\lambda_2}{\lambda_1 + t\lambda_2}$

(b) For simplicity, assume that $\lambda_1 = \lambda_2 = 1$, and let $T = \frac{Y}{X}$.

Use part (a) to find the pdf of T , $f_T(t)$.